AI Literacy

Module 0: Foundations

THE MARCY LAB SCHOOL

By the End of Today, You'll Be Able To...



Explain verified fact vs. AI claim



Name 2 real risks of AI output



State your own double-check rule

CHECK FOR UNDERSTANDING

Quick recall: Is a chatbot's reply structured or unstructured?

A. Structured
B. Unstructured

Answer Key: Chatbot Reply: Type Recall

Correct answer: Unstructured

Free-flowing text with no fixed columns to compare answer to answer — the same judgment call from the structured/unstructured lecture, now applied to something you'll interact with constantly: an AI chat response.

The Goal

Why This Matters

Knowing AI can be wrong is one thing — seeing what actually happens when nobody checks is another.

CASE STUDY

A Court Filing Cited Six Fake Cases

- A lawyer used AI for legal research
- The AI made up fake, confident citations
- None of the six cases existed

Discussion prompt: What single habit would have caught this before it was filed?

Source: Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y. 2023)

A Confident Friend Who's Sometimes Wrong

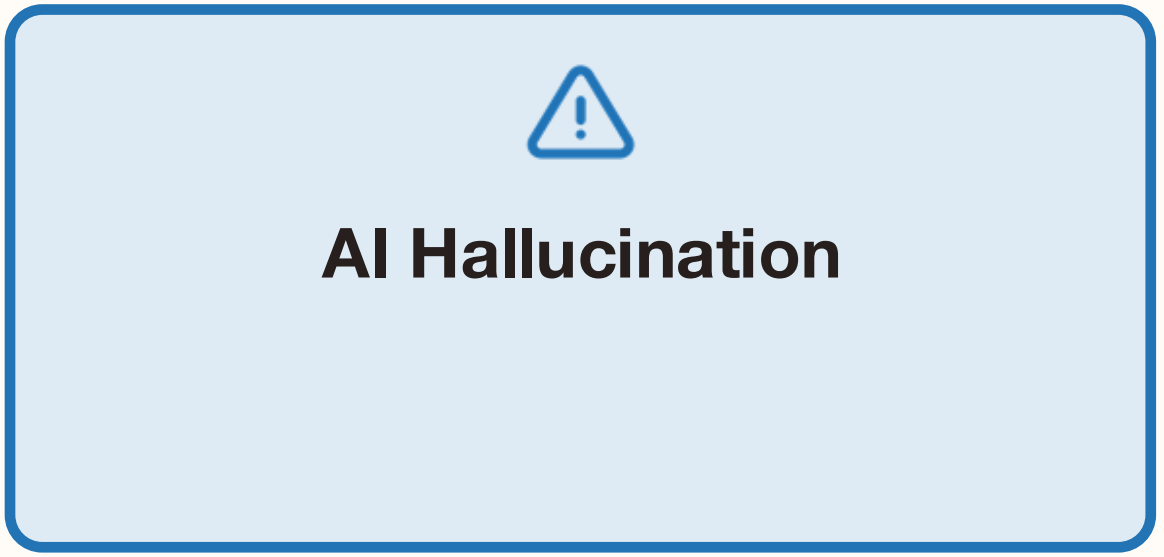
- AI answers sound equally confident either way
- Confidence in tone isn't evidence
- This pattern is called an AI hallucination
- A verified fact survives being checked



Analogy: An AI assistant is like a friend who answers every question fast and sounds completely sure — even when they're guessing. A good friend like this is genuinely useful, but you'd still double-check before repeating something important they told you.

Key Terms





A Chatbot Predicts Words — Never Looks Anything Up



Predicts likely next words



Not a lookup — a fluent guess



Other AI (vision, robots) works differently



Fluent and correct are different things

Understanding the Risk

The Worked Example



Risk 1: A wrong fact, stated confidently



Risk 2: A source that doesn't exist



My rule: Verify before acting or repeating

Worked Example: Naming Real Risks



Risk 1: A wrong fact, stated confidently



Risk 2: A source that doesn't exist



My rule: Verify before acting or repeating

Before You Run an AI-Suggested Command

bash

```
# an AI assistant suggested this command - READ
# it fully and understand it BEFORE running it
$ rm -rf old_data/
```

That One, You Just Read

Code Along: Check Before You Trust

Not every AI-suggested command is dangerous — the habit is reading it either way, so let's actually type one together.

A Safe One: Check Before You Trust

bash

```
# a SAFE AI-suggested command - read-only,
# nothing deleted or changed
$ wc -l data/energy_sample.csv
```

Verify Against the Real File

bash

```
# if an AI told you this file has 100 rows,
# here's how you'd actually check
$ wc -l data/energy_sample.csv
$ head -n 1 data/energy_sample.csv
```

CHECK FOR UNDERSTANDING

An AI assistant gives you a statistic with full confidence but no source. What do you do?

A. Trust it — it sounded confident
B. Ask the AI if it's sure
C. Verify it against a real source before using it

Answer Key: Unsourced Statistic: What Now?

Correct answer: Verify it against a real source before using it

Asking the AI to double-check itself doesn't help — it can restate the same wrong claim just as confidently. A real external source is the only real check.

WOULD YOU RATHER?

Would you rather an AI assistant say "I'm not sure" honestly, or guess confidently and sometimes be wrong?

QUICK CHALLENGE

Round-robin: Name one thing you'd never trust an AI's answer on without double-checking.

A. A statistic or number
B. A quote or citation
C. A command that deletes or changes files
D. A legal or medical claim

Answer Key: Never Trust AI On...

Correct answer: Depends on the item named live

All four are real, common categories where AI tools are known to confidently state wrong information — a strong personal rule usually covers at least one of these explicitly.

TURN AND TALK

In pairs: Name your own 2 concrete AI-output risks and one personal double-check rule — not the ones from the worked example.

TURN AND TALK

Discussion: What's one time (in class or outside it) an AI tool gave you something that turned out wrong?

QUICK CHALLENGE

Round-robin: What's a real, checkable source you could use to verify a claim about a dataset?

A. Another AI's answer to the same question
B. The publisher's own official page
C. A government or research-institution site

Answer Key: Name a Checkable Source

Correct answer: The publisher's own official page

The publisher's own page (or a government/research source) is a real, checkable source. A second AI's answer isn't independent verification — it's the same underlying risk again.

Key Terms — Recap

Term	Definition
Verified fact	Something confirmed true against a real, checkable source
AI hallucination	A confident-sounding AI answer that is factually wrong

The Practice

Wrapping Up

Checking one AI-suggested command today is the same habit you'll need every time AI hands you an answer, not just here.

What You Have Now

- What an AI hallucination really means
- A worked example: Risks plus a rule
- Practice applying that rule live, today

Before Next Class

- Next class: Lab, column-typing practice
- After that: Project launch and full brief

KEEP LEARNING

Where to Go From Here

📖 **AI Hallucinations, Explained (IBM)**
IBM · READING · 8-10 min
A clear, current explainer on why this happens.

>_ **Google: AI Literacy Fundamentals**
Google · PRACTICE · 15-20 min, self-paced
Free, interactive modules on using AI tools well.